Unit 3, Lesson 7: Interpreting Linear Models

Benchmark

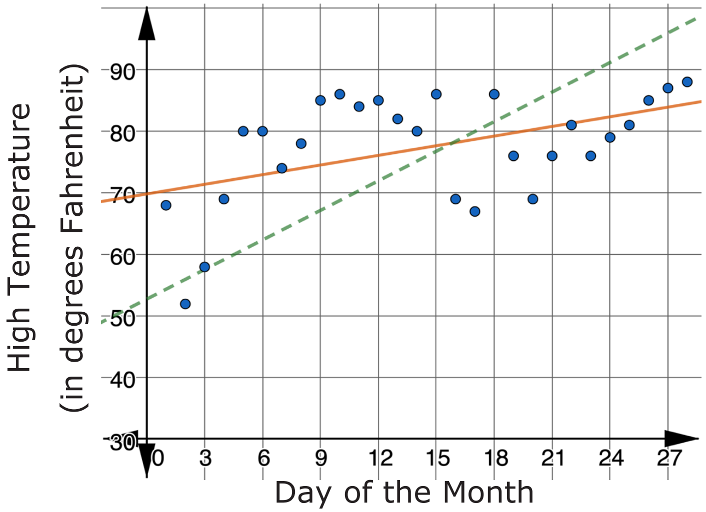
MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

Learning Targets

- ☐ I can interpret the correlation coefficient, slope, and y-intercept of a linear equation fit to real-world data.
- ☐ I can use the linear equation fit to bivariate data to solve real-world problems in terms of the context of the data.

3.7.1 Warm-Up: Bell Work

1. The scatter plot shows the relationship between the day of the month and the high temperature, in degrees Fahrenheit, for February 2021 in Brooksville, FL.



a. Which line fits the data better, the solid orange line or the dashed green line?

b. Explain your reasoning.

c. Use inequalities to represent an appropriate domain and range for the data represented.

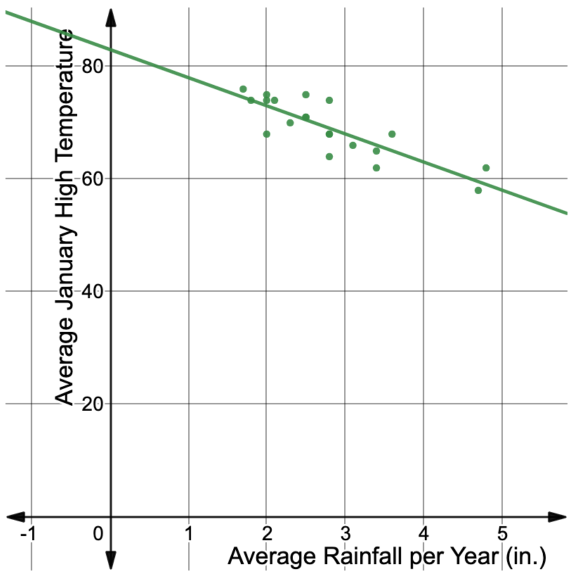
Domain	
Range	

3.7.2 Activity: Station 1: High Temperature and Rainfall

The table shows the mean high temperature in January, in degrees Fahrenheit ($^\circ F$), and the mean amount of rainfall per year, in inches (in.), for 20 cities in Florida.

City	Mean Rain per Year (in.)	Mean High Temp in Jan. ($^\circ F$)	City	Mean Rain per Year (in.)	Mean High Temp in Jan. ($^\circ F$)
Alachua	3.6	68	Naples	1.7	76
Bradenton	2.5	71	Orlando	2.3	70
Clermont	2.8	68	Pensacola	4.7	58
Fort Lauderdale	2.5	75	Sarasota	2.5	71
Fort Myers	1.8	74	St. Augustine	3.1	66
Fort Pierce	2.1	74	Tallahassee	4.8	62
Gainesville	3.4	65	Tampa	2.0	68
Jupiter	2.8	64	West Palm Beach	2.8	74
Key West	2.0	74	Winter Garden	2.8	68
Miami	2.0	75	Yulee	3.4	62

The scatter plot shown represents the data from the table with a linear equation fit to model the relationship.



1. The screenshot shows the slope, y-intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS

$r^2 = 0.7036$

$r = -0.8388$

RESIDUALS

e_1 plot

PARAMETERS

$m = -4.99154 \quad b = 83.0265$

a. Write the equation of the line of best fit shown.

b. Interpret the meaning of each value listed in the table.

Value	Meaning
slope, m	
y-intercept, b	
correlation coefficient, r	

2. Use inequalities to express an appropriate domain and range using the linear equation that models the data.

Domain	
Range	

3. Use the model to predict the mean high temperature in January for a Florida city that has a mean of 4.0 inches of rainfall per year.

4. Suppose there is a city in Florida that averages 7.8 inches of rainfall per year.

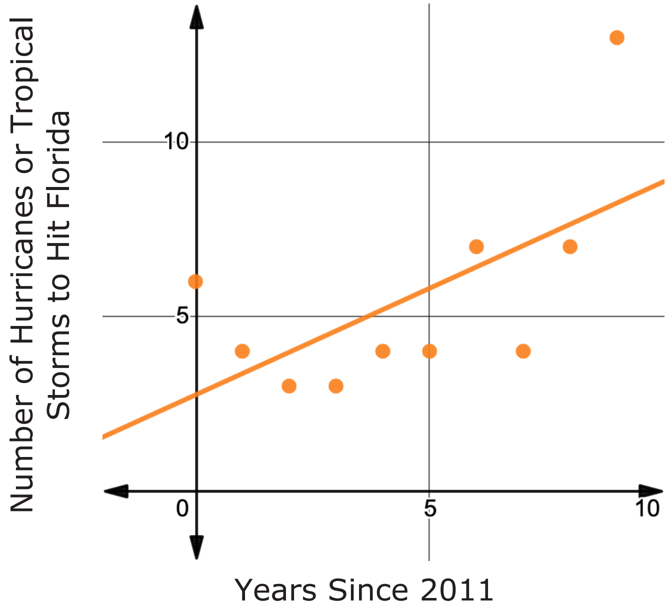
   a. Use the equation to predict the mean high temperature in January for that city.

   b. Explain whether or not this January high temperature seems reasonable.

   c. Discuss with your partner if you think the model can be extrapolated to a city in Florida averaging 7.8 inches of rainfall per year. Summarize your discussion.

3.7.3 Activity: Station 2: Hurricanes in Florida Since 2011

Michaela wondered if there was a trend between the number of hurricanes and tropical storms that hit Florida and time. She used data for the number of hurricanes and tropical storms to hit Florida from 2011-2020 to create the scatter plot shown.



1. The equation $y = 0.612x + 2.75$ models this relationship.

   a. Interpret the meaning of the slope and y-intercept in context.

Value	Meaning
slope, m	
y-intercept, b	

   b. Explain why it does not make sense for the slope or y-intercept to be decimal values in this context.

2. Use the model for the relationship, $y = 0.612x + 2.75$, to complete the following.

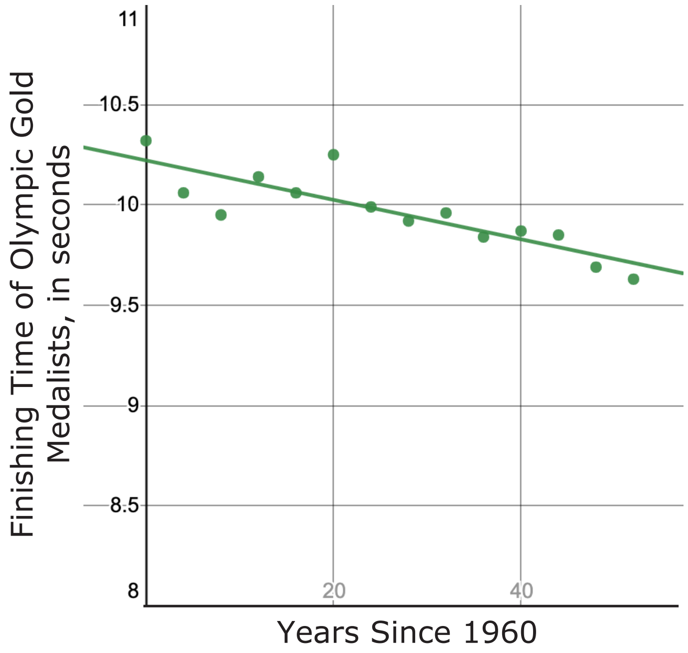
   a. Predict the number of hurricanes and tropical storms expected to hit Florida in 2023.

   b. Explain whether or not it seems reasonable for the number of hurricanes and tropical storms to hit Florida to continue increasing at this rate each year.

   c. Discuss with your partner how the model could be used to estimate the number of hurricanes and tropical storms that hit Florida in 2008. Summarize your discussion.

3.7.4 Activity: Station 3: Men's 100-Meter Dash at the Olympics

The scatter plot shows the finishing times for the Olympic gold medalist in the men's 100-meter (m.) dash for the Olympic games since 1960.



1. Dana says that the strength and direction of the linear association between the two variables is moderately negative. Explain whether or not you agree with Dana.

2. The equation of the line of best fit shown on the scatter plot is $f(x) = 10.2217 - 0.0098x$, where f is the finishing time of the Olympic gold medalist in the men's 100-m dash and x is the number of years since 1960.

   a. Interpret the y-intercept of the linear equation modeling the data.

   b. Discuss with your partner the difference between the finishing time for the men's gold medal winner for the 100-m dash in the 1960 Olympics and the y-intercept of the linear equation.

   c. Interpret the slope of the linear equation that models the data.

d. The summer Olympic games, where the 100-m dash is run, occur every 4 years. By how many seconds does the model predict the first-place finishing time will decrease every 4 years?

e. Discuss with your partner whether or not it is reasonable to use this model to predict the first place finishing time in the men's 100-m dash for the 2050 summer Olympics. Summarize your discussion.

3.7.5 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

My favorite mistake from today was ...